Recall Modules over a ring R (left), homomorphisms, isomorphisms, submodules, quotients.

R field $\Longrightarrow$ vector space.

§ Direct product and direct sum

Definition R ring, $(M_i)_{i\in I}$ a family of R-modules, $\prod_{i\in I} M_i$ defined as an additive group.

Make it into an R-module:

$$\begin{aligned} R \times \prod_{i\in I} M_i &\longrightarrow \prod_{i\in I} M_i \\ \left(r, (x_i)_{i\in I}\right) &\longmapsto (rx_i)_{i\in I} \end{aligned}$$

where $\prod_{i\in I} M_i$ is called the **direct product**.

$$\bigoplus_{i\in I} M_i := \left\{(x_i)_{i\in I} \mid \exists I_0 \in I \text{ finite}, \forall i \in I \setminus I_0, x_i = 0\right\}$$

is called the **direct sum**.

Convention $I = \emptyset \Longrightarrow \prod = \bigoplus = \{0\}$. Notations are similar to vector spaces.

$\forall j \in I$, we have homomorphisms $\iota_j : M_j \hookrightarrow \bigoplus_{i\in I} M_i$, $p_j : \bigoplus_{i\in I} M_i \twoheadrightarrow M_j \Longrightarrow$ Can identify M_j as a submodule of $\bigoplus_{i\in I} M_i$. Then $\bigoplus M_i = \sum$ of these submodules.

Internal version

Definition-Proposition M R-module, $(M_i)_{i\in I}$ family of submodules.
Define $\sigma : \bigoplus_{i\in I} M_i \to M$, $(x_i)_{i\in I} \mapsto \sum x_i$. The following are equivalent:
(1) $\sum M_i = M$, and $\forall i \in I, M_i \cap \sum_{j\neq i} M_j = \{0\}$; (2) $\forall x \in M, \exists! (x_i)_{i\in I}$ s.t. $x = \sum x_i$;
(3) σ is an isomorphism.

If these hold, we say M is the (internal) direct sum of submodules $(M_i)_{i\in I}$.

Proof Same as the case of vector spaces. □

§ Free module

Definition X set, $R^{\oplus X}$ R-module, along with map $X \to R^{\oplus X}, x \mapsto \left(\delta_{xy}\right)_{y\in X}$. It is called the **free module** with basis X.

Observation $\forall R$-module $N, \mathrm{Hom}_R(R^{\oplus X}, N) \longleftrightarrow \{\text{maps } X \to N\}$ where

$$\begin{aligned} \varphi &\longmapsto \varphi_X, \\ \left[\varphi\left((r_x)_x\right) = \sum_x r_x f(x)\right] &\longleftarrow (f : X \to N). \end{aligned}$$

Question Given M, subset $X \subset M$, does these exist $M \rightsquigarrow R^{\oplus X}$?

Definition-Proposition The following are equivalent:

- X generates M, and X is linearly independent, i.e., $\sum_{x\in X} r_x x = 0 \iff \forall x, r_x = 0$.
- $\forall y \in M, \exists!(r_x)_{x\in X} \in R^{\oplus X}$ s.t. $y = \sum_{x\in X} r_x x$.
- Let $\varphi : R^{\oplus X} \to M$ be the homomorphism corresponding to $X \hookrightarrow M, \varphi$ is an isomorphism.

In this case, we say M is free with basis X. We say an R-module M is free if $\exists X \subset M$ s.t. M is free with basis X. ($\iff \exists$ isomorphism $R^{\oplus X} \xrightarrow{\sim} M$ for some X)

Proof (1) $\iff$ (2): X generates $M \iff$ existence of $(r_x)_x$. Linearly independent $\iff$ unique.

(2) $\iff$ (3): Existence $\iff \varphi$ surjective, uniqueness $\iff \varphi$ injective. □

Remark R field, then $\forall R$-module is free! (also true for division ring, not true in general)

Example $R = \mathbb{Z}, M := \mathbb{Z}/n\mathbb{Z}, n \notin \{0, \pm 1\} \implies M \not\simeq \mathbb{Z}^{\oplus X}$ for any set X by counting.

Definition-Proposition R commutative ring, M free with basis X, Y, then $|X| = |Y|$. Hence we can define its **rank** as $\mathrm{rk}(M) := |X|$ for any basis X of M.

Proof (the case R integral domain, M finitely generated)

First: $\forall$ basis $X, |X| < \infty, \because$ Identify M with $R^{\oplus X}$. But X infinite $\iff R^{\oplus X}$ not finitely generated!

Secondly $|Y| \le |X| =: n$. Identify M with $R^{\oplus n} \hookrightarrow K^{\oplus n}$ where $K := \mathrm{Frac}(R)$.
$\forall y_1, ..., y_{n+1}$ in $M = R^{\oplus n}, \exists a_1, ..., a_{n+1} \in K$ not all 0 s.t. $\sum_{i=1}^{n+1} a_i y_i = 0$ in K^n : K-vector space.
May assume $\forall i, a_i \in R$, hence $y_1, ..., y_{n+1}$ are linearly dependent in $M, |Y| \le n$. □

Remark $\exists$ version for right R-modules. Fact: (right modules)

$$\begin{array}{ccc} \mathrm{Hom}_R(R^{\oplus n}, R^{\oplus m}) & \longleftrightarrow & M_{m\times n}(R) \\ \left[\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right] & \longleftarrow\!\!\mid & A \end{array}$$

we use right module to ensure for $t \in R$,

$$\begin{pmatrix} x_1 t \\ \vdots \\ x_n t \end{pmatrix} \mapsto \left[A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right] t.$$

For left modules: should use row vectors and right multiplication.

Key example F field, X variable. M $F[X]$-module $\longrightarrow$ get F-vector space structure on M via $F \hookrightarrow F[X]$ as subring.

Lemma V F-vector space, upgrade V to an $F[X]$-module $\longleftrightarrow$ prescribe $T \in \mathrm{End}_F(V)$ by letting $f \in F[X]$ act on V via $f(T) \in \mathrm{End}_F(V)$. Furthermore, given (V, T) and (V', T'),

$$\mathrm{Hom}_{F[X]}(V, V') = \{\varphi \in \mathrm{Hom}_F(V, V') \mid \varphi T = T'\varphi\}.$$

Proof Upgrade $V \longleftrightarrow$ prescribe $\forall f \in F[X]$ with $\rho_f : V \to V$ s.t.

$$\rho_f(v_1 + v_2) = \rho_f v_1 + \rho_f v_2, \quad \rho_{f+g} = \rho_f + \rho_g, \quad \rho_{fg} = \rho_f \rho_g, \quad \rho_t(v) = tv, \forall t \in F.$$

$\implies \rho_f(v) = f(T)v$ if $T := \rho(X) : V \to V$ is F-linear.

Conversly, given T, can define $\rho_f = f(T)$ to make V into $F[X]$-module.

Given $V, V', \varphi \in \mathrm{Hom}_F(V, V')$,

$$\begin{aligned}\varphi \in \mathrm{Hom}_F(V, V') &\Longleftrightarrow \varphi\rho_f = \rho'_f\varphi, \forall f \in F[X] \\ &\Longleftrightarrow \varphi\rho_X = \rho'_X\varphi \Longleftrightarrow \varphi T = T'\varphi.\end{aligned}$$

□

Corollary Given $(V, T), (V', T')$, then

$$\{\text{isom } \varphi : V \to V' \text{ as } F[X]\text{-modules}\} = \{F\text{-linear } \varphi : V \xrightarrow{\sim} V' \text{ s.t. } \varphi T = T'\varphi\}.$$

Goal Classify all $F[X]$-modules V (or data (V, T)) up to $\simeq$, where $\dim_F V =: n < \infty$.

Define $(V, T) \approx (V', T')$ if $\exists \varphi : V \to V'$ F-linear s.t. $\varphi T = T'\varphi$.

$$\begin{aligned}&\{V : F[X]\text{-module}, \dim_F V = n\}/\simeq \\ \longleftrightarrow\ &\{(V, T) : T \in \mathrm{End}_F(V), \dim_F V = n\}/\approx \\ \longleftrightarrow\ &M_{n\times n}(F)/\text{ conjugation } (V = F^n)\end{aligned}$$

Key: $F[X]$ is a PID, and $\dim_F V < \infty \Longrightarrow V$ is a finitely generated $F[X]$-module.

Theorem R PID, M finitely generated R-module, then

$$M \simeq R/I_1 \oplus \ldots \oplus R/I_k \oplus E$$

where

- $I_1 \supset \ldots \supset I_k$: non-trivial ideals in R,
- $k \in \mathbb{Z}_{\geq 0}$,
- E finitely generated free R-module, and if

  $$R/I_1 \oplus \ldots \oplus R/I_k \oplus E \simeq R/I'_1 \oplus \ldots \oplus R/I'_k \oplus E',$$

  then
  - $k = k', \forall i, I_i = I'_i$,
  - $E \simeq E'$ ($\Longleftrightarrow \mathrm{rk}(E) = \mathrm{rk}(E')$).

The sequence of ideals $I_1 \supset \ldots I_k \supset \{0\} \supset \ldots \supset \{0\}$ is called the invariant factors / 不变因子 of M.

$R = F[X], M = F[X]/(f), f = c_0 + \ldots + c_{n-1}X^{n-1} + X^n$.
Take the basis of $M : 1 + (f), \ldots, X^{n-1} + (f)$, then

$$\begin{aligned}T(X^i + (f)) &= x^{i+1} + (f), \\ T(X^{n-1} + (f)) &= -c_0 - \ldots - c_{n-1}X^{n-1} + (f),\end{aligned}$$

hence the matrix of T is

$$c_f = \begin{pmatrix} 0 & 0 & \ldots & 0 & -c_0 \\ 1 & 0 & \ldots & 0 & -c_1 \\ 0 & 1 & \ldots & 0 & -c_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \ldots & 1 & -c_{n-1} \end{pmatrix}$$